$(2p+1) \bmod (2n+1) = p'$.

$2p' + 2(2n+1) \cdot \square = 2(2p+1)$

$(2p'+1) \bmod (2n+1) = p''$ $\qquad 2^k(p+1) - 1$

$[2(2p+1) - 2(2n+1) \cdot 2 + 1] \bmod (2n+1) = p''$.

$[4p+3] \neq \bmod (2n+1)$

$\left(2[2(2p+1)+1]+1 \cdots \right) \bmod (2n+1)$

$2^k p + 2^k - 1$.

1 3 7 15 31 63

$2a_n + 1 = a_{n+1}$.

$2^n - 1$ $\qquad 2 \cdot 2^n - 1 = 2^{n+1} - 1$

$$\left[2^k(p+1)-1\right] \bmod (2n+1) = p.$$

$$(2n+1)z + p = 2^k(p+1)-1$$

$$53z = 2^k - 1.$$